Doping Stability Report -- Cr in KCoO2 at 300 C

Generated: 2026-05-30 12:46   Host: KCoO2   Dopant: Cr

Executive Summary

Cr preferentially occupies the Co layer. The formation energy difference between the two layers is +6.212 eV, strongly favouring the Co site. MD confirms this preference: on the Co site the dopant MSD plateau slope is -0.00713 A^2/ps (stable), compared to 0.00930 A^2/ps on the K site. The Co-site result is isovalent (no charge mismatch), so the formation energy is fully reliable. The K-site E_f is approximate due to a charge surplus of +2 that cannot be modelled in CHGNet.

Site	E_f (eV)	Charge	Compensation	MSD pass	Coord pass	Verdict
Co	-3.428	+0	No	PASS	PASS	STABLE
K	+2.785	+2	Yes	FAIL	PASS	MIGRATION / UNFAVORABLE

Cr at the Co layer -- STABLE

Charge situation

Cr replaces Co with no change in oxidation state (isovalent substitution, charge mismatch = 0). No charge compensation is needed. The formation energy is fully reliable.

Formation Energy (thermodynamic stability)

Formation energy E_f	**-3.4277 eV**	PASS	*< 1.0 eV to pass*

E_f = -3.428 eV. This value is exceptionally large and negative, indicating that Cr incorporation is strongly thermodynamically driven at the Co site.

Molecular Dynamics Stability (300 C, 25 ps)

MSD slope (late-time)	**-0.00713 A^2/ps**	PASS	*< 0.005 to pass (plateau = stable)*
Final MSD	**0.207 A^2**	----	
Max displacement	**0.954 A**	----	
Coordination (min/mean/max)	**4 / 4.5 / 5**	PASS	*relaxed = 5, tolerance +/-1*
Mean bond length (MD vs relaxed)	**1.809 A (relaxed: 1.8**	----	
Lattice volume change	**+0.00 %**	PASS	*< +/-5% to pass*

Late-time MSD slope = -0.00713 A^2/ps. The negative slope is a flat plateau -- essentially no migration detected. Final MSD = 0.207 A^2; maximum displacement from starting position = 0.95 A.

Coordination fluctuated between 4 and 5 (mean 4.5), within the +/-1 tolerance of the relaxed value (5). This is consistent with normal thermal vibration. Mean nearest-neighbour distance during MD = 1.809 A.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: STABLE

Cr is thermodynamically favorable and kinetically stable on the Co site at 300 C. All four stability criteria pass: formation energy is favorable, the dopant does not migrate during MD, its coordination environment is preserved, and the host lattice volume is unchanged.

Cr at the K layer -- MIGRATION / UNFAVORABLE

Charge situation

Cr replaces K with a charge surplus of +2. CHGNet cannot model this compensation (would require Co oxidation or electron addition). The formation energy below is approximate.

Formation Energy (thermodynamic stability)

Formation energy E_f	**+2.7847 eV**	FAIL	*< 1.0 eV to pass*

E_f = +2.785 eV (approximate -- charge surplus not modelled in CHGNet). This value is large and positive, indicating that Cr incorporation is thermodynamically unfavorable under equilibrium conditions at the K site.

Molecular Dynamics Stability (300 C, 25 ps)

MSD slope (late-time)	**0.00930 A^2/ps**	FAIL	*< 0.005 to pass (plateau = stable)*
Final MSD	**0.510 A^2**	----	
Max displacement	**1.308 A**	----	
Coordination (min/mean/max)	**4 / 4.0 / 5**	PASS	*relaxed = 5, tolerance +/-1*
Mean bond length (MD vs relaxed)	**1.692 A (relaxed: 1.9**	----	
Lattice volume change	**+0.00 %**	PASS	*< +/-5% to pass*

Late-time MSD slope = 0.00930 A^2/ps. The slope exceeds the plateau threshold, suggesting mild drift or site-hopping. Final MSD = 0.510 A^2; maximum displacement from starting position = 1.31 A.

Coordination fluctuated between 4 and 5 (mean 4.0), within the +/-1 tolerance of the relaxed value (5). This is consistent with normal thermal vibration. Mean nearest-neighbour distance during MD = 1.692 A.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: MIGRATION / UNFAVORABLE

The formation energy is favorable but Cr migrates away from the K site during MD. The relaxed structure is not the true resting site. Inspect the MSD and trajectory to identify where the dopant ends up.

Site Preference Comparison

The Co site is preferred by 6.212 eV over the K site. Formation energy: -3.428 eV (Co) vs +2.785 eV (K).

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Caveats and Limitations

? CHGNet is charge-neutral.

The potential does not model oxidation states, polaron formation, or electron/hole localisation. Formation energies for aliovalent dopants (charge mismatch != 0) are approximate even when structural compensation defects are included.

? **No Freysoldt/Kumagai correction.**

Electrostatic finite-size corrections for charged defects require the host dielectric tensor and the DFT electrostatic potential, neither of which is available from a neural-network potential. For publication-quality E_f of charged defects, follow up with DFT using doped or pymatgen-analysis-defects.

? **Metal-rich reference state.**

Chemical potentials are computed from elemental ground-state structures. Synthesis under oxide-rich or other conditions shifts all E_f values by additive constants; the relative ordering of sites is unaffected.

? **MD timescales are short (25 ps default).**

This is sufficient to assess local site stability and rapid migration, but cannot resolve slow diffusion mechanisms. A non-plateauing MSD confirms instability; a plateau does not guarantee stability on longer timescales.

? **Energy rankings are more trustworthy than absolute values.**

CHGNet is a universal MLIP trained on r2SCAN DFT. Formation-energy errors for transition-metal oxides are typically 50-150 meV/atom. Use relative orderings (which site is preferred?) as the primary result.